Application for Capstone Project (W25/26)

Last Name: Hullukatte Nataraju

First Name: Venushambhu

Matriculation No: 82059298

Study Program

Masters in Data Science

Title of Project

Unified Financial Risk Intelligence Platform

Short Description (2-3 Paragraphs)

The Unified Financial Risk Intelligence Platform is a Data Science based system designed to analyze and predict multiple types of financial risks like credit risk, fraud risk, and market risk using machine learning and predictive analytics. The platform integrates data from various financial sources, builds separate risk models, and combines their outputs into a Unified Risk Index (URI) that provides a holistic view of an institution's overall financial exposure. An interactive dashboard will visualize these insights, enabling banks and fintech firms to make more informed and data-driven decisions.

- [x] I am aware of the workload requirements for this module
- [ ] I will conduct this project together with an outside partner
- [x] I am opening the project for potential teamwork
- [ ] The project is under an NDA

Place, Date: Berlin,19.10.2025

Signature: venushambhu

For internal use

Project approved

Project proposal requires revision

Place, Date: Berlin ,19.10.2025

Signature: Venushambhu

Application for Capstone Project (W25/26)

Last Name: Hulikere Nagaraju

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Matriculation No: 38966504

Study Program

Master ’ s in Data Science

Title of Project

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Capstone Project Proposal

Unified Financial Risk Intelligence Platform

1. Project Statement

The proposed project aims to design and develop a data science–driven financial risk intelligence platform that integrates predictive analytics for credit risk, fraud risk, and market risk under one unified analytical system. By leveraging machine learning, anomaly detection, and time-series forecasting, the platform will identify and quantify potential financial exposures. The system's output, the Unified Risk Index (URI), will help financial institutions and fintech organizations make smarter, data-backed decisions to mitigate losses and enhance operational stability. This project emphasizes modularity, transparency, and interpretability of results.

2. Core Project Objectives

1. **Develop a Credit Risk Prediction Model:** Build and train supervised learning models (e.g., Logistic Regression, Random Forest, XGBoost) to predict loan default probability using credit and demographic data.
2. **Implement a Fraud Detection Module:** Use unsupervised learning techniques (Isolation Forest , Autoencoders) to detect abnormal transaction behaviors that could indicate fraudulent activity.
3. **Build a Market Risk Forecasting Model:** Apply time series forecasting models (ARIMA, Prophet, LSTM) to predict market volatility and portfolio risk using financial market data.
4. **Design a Unified Risk Scoring Framework:** Combine outputs from the above models into a Unified Risk Index (URI) representing overall financial exposure for customers or institutions.
5. **Develop an Interactive Dashboard:** Create a **Streamlit** that visualizes risk scores, alerts, and performance trends for improved managerial decision-making.

3. Technical Implementation Plan

3.1. Data Sourcing and Preprocessing Pipeline

- **Data Collection:**
  - Credit Risk: *Home Credit Default Risk* or *Lending Club Loan Data* (Kaggle)
  - Fraud Risk: *Credit Card Fraud Detection Dataset* (Kaggle)
  - Market Risk: Stock/portfolio data from *Yahoo Finance API*
- **Data Preprocessing:**
  - Handle missing values and normalize financial features.
  - Encode categorical attributes (e.g., income type, loan type).
  - Feature engineering: create new indicators (debt-to-income ratio, transaction frequency).
  - Merge datasets into a unified schema representing customer and market profiles.

3.2. Model Architecture and Training

- **Credit Risk Model:**
  - Train supervised ML models (Random Forest, XGBoost).
  - Evaluate using AUC, F1-score, and precision-recall metrics.
- **Fraud Detection Model:**
  - Implement anomaly detection (Isolation Forest, Autoencoder).
  - Evaluate using confusion matrix and ROC-AUC for imbalanced data.
- **Market Risk Model:**
  - Use ARIMA or Prophet to forecast stock volatility and trends.
  - Evaluate using RMSE and Mean Absolute Error (MAE).
- **Unified Risk Index (URI):**
  - Compute a weighted aggregate:

    $$\mathbf{URI=0.4\times CreditRisk+0.4\times FraudRisk+0.2\times MarketRisk}$$

  - Normalize scores to produce interpretable risk categories (Low, Medium, High).

3.3. Visualization & Dashboard Development

- Build an interactive Streamlit dashboard to display insights, trends, and risk distributions across multiple financial dimensions.
- **Tools**: **Streamlit** for real-time visualization.

4. Evaluation Plan

Model Type	Evaluation Metrics	Goal
Credit Risk Model	AUC, F1-Score, Precision-Recall	≥ 85% predictive accuracy
Fraud Detection	ROC-AUC, Confusion Matrix	Detect ≥ 90% of anomalies
Market Risk Model	RMSE, MAE	Accurate volatility forecasting
Unified Risk Score	Cross-validation & interpretability	Reliable composite risk scoring

A qualitative evaluation will include visual demonstrations of the risk dashboard and interpretation of model outputs.

5. Technology Stack

- **Programming Language:** Python
- **ML Frameworks:** Scikit-learn, XGBoost, TensorFlow
- **Data Handling:** Pandas, NumPy, SQL
- **Model Evaluation & Explainability:** AUC, F1-Score, Precision-Recall MLflow (tracking), Cross-Validation, ROC-AUC, Confusion Matrix
- **Time Series Analysis:** Statsmodels, Prophet,ARIMA
- **Visualization:** Streamlit, Plotly,
- **Data Sources:** Kaggle, Yahoo Finance API
- **Version Control:** Git / GitHub

6. Project Timeline & Milestones

Milestone	Duration	Target Completion
Sprint 1: Data Sourcing & Cleaning	2 Weeks	30 Oct 2025
Sprint 2: Credit Risk Model Development	2 Weeks	13 Nov 2025
Sprint 3: Fraud Detection & Market Risk Models	2 Weeks	27 Nov 2025
Sprint 4: Unified Risk Index & Dashboard Integration	2 Weeks	11 Dec 2025
Sprint 5: Evaluation, Deployment & Documentation	2 Weeks	25 Dec 2025

7. Expected Deliverables

- Cleaned and integrated financial datasets.
- Trained ML models for credit, fraud, and market risk.
- Unified Risk Index computation framework.
- Interactive visualization dashboard.
- Final report and demonstration notebook.

8. Expected Impact

The Unified Financial Risk Intelligence Platform will enable banks and fintech institutions to predict risk proactively, minimize financial losses, and improve credit and transaction decision-making through unified, explainable, and data-driven analytics.